

Date: 7th April 2017	Detailed Design Document	
Revision: 1.2.2		
Resp: ARM Teamlead		
Page: 1 of 10		

TRW Automotive

Slip Control Software Design

Document ID	:	ARM-DDD-Arm_State
Title	:	ARM-DDD-Arm_State
Summary	:	Description of subsystem architecture and interface
Scope	:	Application: arm_state{.c, _sub.h} Calibration:
Doc Template	:	DDD-Module-Template based on Template C43-03-006-SCS-EN-F01 Issue B
Project Name	:	AR 16 Core Development
Doc Status	:	Draft / For Review / Final / Released
Audience	:	Only for TRW internal use.

Release Date	Author	Reviewer	Approved by
	A. Hoehn		

REFERENCES

Ref.	Document	Title	Ver.	Location

REVISION HISTORY

Rev.	Date dd/mm/yyyy	Author	Change Request	Comments (Version)
1.0	30/05/2011	B. Etges		Initial Version
1.2	23/08/2016	A. Hoehn		Update to AR 16
1.2.1	20/12/2016	A. Hoehn		Activation Image
1.2.2	07/04/2017	A. Hoehn		Ay_Filt_Reset

Contents

1	ABSTRACT	3
2	ACRONYMS	3
3	OVERVIEW	3
3.1	DESIGN GOAL	3
3.2	GENERAL REQUIREMENTS	3
3.3	DETAILED REQUIREMENTS	3
3.4	SYSTEM MODE REQUIREMENTS	4
3.5	FLOW-CHARTS AND DIAGRAMS	4
3.5.1	Overview Arm Trigger Conditions	4
4	DETAILED DESIGN	4
4.1	ALGORITHM DESCRIPTION	4
4.1.1	ARM Activation	4
4.1.2	ARM Deactivation	6
4.1.3	Vehicle Turn Direction	7
5	PROVIDED TASKS	7
5.1	ArmState	7
6	PROVIDED SERVICES	7
6.0.1	Private Services	8
7	CONFIGURATION OPTIONS	8
8	CONFIGURATION PROCEDURES	8
9	TEST CASES	8
10	CALIBRATION INFORMATION	8
10.1	Activation	8
10.1.1	Vehicle Speed	8
10.1.2	Roll Index Flag	9
10.1.3	Rollover Criticality	9
10.1.4	YawRate Condition	9
10.2	Deactivation	10
10.2.1	Ay_Filt_Reset	10
11	APPENDIX	10

1 ABSTRACT

This module controls the activation and deactivation of the Active Roll Mitigation ([ARM](#)).

2 ACRONYMS

ARM Active Roll Mitigation

ROI Roll Over Index

SWA Steering Wheel Angle

AyMod 'Modified' Lateral Acceleration, that tries to predict the Lateral Acceleration

EoL End of Line

3 OVERVIEW

This document describes the purpose and behavior of Arm_State.

3.1 DESIGN GOAL

[ARM](#) observes the vehicle behavior and the driver input in order to detect critical situations which indicate a impending roll over .

3.2 GENERAL REQUIREMENTS

1. [ARM](#) should prevent dynamic rollover.
2. [ARM](#) should prevent static rollover.
3. [ARM](#) should be robust so that the reason for each significant ARM intervention is comprehensible to the end customer.

3.3 DETAILED REQUIREMENTS

Requirement ID	
Location	

3.4 SYSTEM MODE REQUIREMENTS

This module needs only to be processed, when **ARM** is allowed.

3.5 FLOW-CHARTS AND DIAGRAMS

3.5.1 Overview Arm Trigger Conditions

Fig. 3.1 describes **ARM** trigger conditions and shows the basic signal flow in **ARM**.

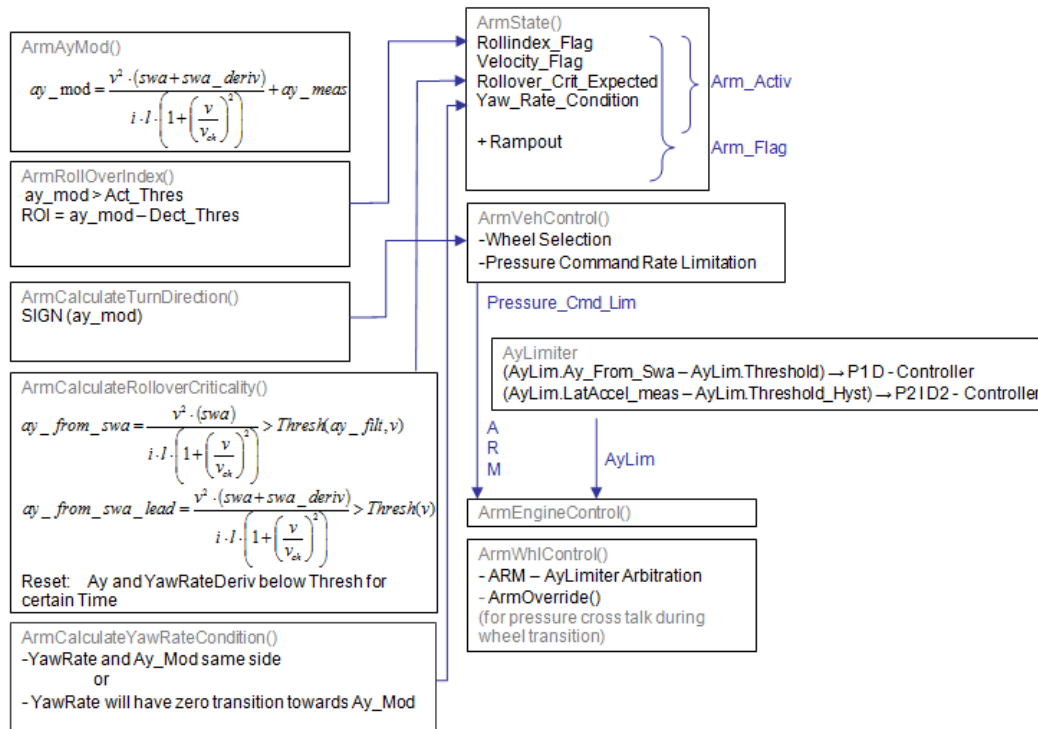


Figure 3.1: **ARM** Overview

4 DETAILED DESIGN

This chapter outlines the detailed design of **Arm_State**.

4.1 ALGORITHM DESCRIPTION

4.1.1 **ARM** Activation

Arm is activated (**Arm_Active** is set), if all of the following conditions are fulfilled:

- **Rollindex_Flag** is set
- **Veh_Velocity_Thr_Flag** is set

- Rollover_Crit_Expected is set
- Yaw_Rate_Condition is set

Rollindex Condition

If the Roll Over Index (ROI) exceeds the activation threshold, the Rollindex_Flag is set. For further information see ARM-DDD_Arm_Roll_Over_Index.pdf.

Velocity Condition

ARM provides a Trim for the minimum vehicle velocity for ARM activation. In addition, a 2Dimensional Lookup table provides a pressure gain depending on vehicle speed. This gain is called Arm_Veh_Control.Speed.Gain. During ARM activation, this gain needs to be unequal to zero, otherwise ARM stays deactivated.

Rollover Criticality

The roll over criticality is calculated in ArmCalculateRolloverCriticality. It is derived from two different lateral accelerations that are calculated using a single track model with steering angle input. One of these is a prediction.

$$Ay_From_Swa = f(swa) \quad (4.1)$$

$$Ay_From_Swa_Lead = f\left(\frac{d}{dt} \cdot swa \cdot preview_time\right) \quad (4.2)$$

Remark: The naming is a bit confusing since Ay_From_Swa_Lead is rather the estimated lateral acceleration of the scaled SWA derivative than a real preview of Ay_From_Swa. The latter would be $Ay_From_Swa + \frac{d}{dt} Ay_From_Swa \cdot TRIM$

The threshold for Ay_From_Swa is depending on vehicle velocity and on the maximum lateral acceleration that occurred in the nearest past.

$$Ay_From_Swa_Thresh = TRIM \cdot f(velocity) \cdot f(Ay_Meas_Hold) \quad (4.3)$$

The threshold for Ay_From_Swa_Lead is only depending on vehicle speed.

$$Ay_From_Swa_Lead_Thresh = TRIM \cdot f(velocity) \quad (4.4)$$

A Rollover critical event is expected and the flag Rollover_Crit_Expected is set, if

$$Ay_From_Swa > Ay_From_Swa_Thresh \quad (4.5)$$

$$Ay_From_Swa_Lead > Ay_From_Swa_Lead_Thresh \quad (4.6)$$

Ay_From_Swa_Lead is very fast, hence a postrun timer is used to capture that Ay_From_Swa_Lead exceeded the Ay_From_Swa_Lead_Thresh threshold.

Within an adjustable time after the Rollover_Crit_Expected flag is set, the measured lateral acceleration or the yaw rate derivative has to exceed their thresholds in order to set Rollover_Crit_Confirmed flag, otherwise the Rollover_Crit_Reset flag is set. This leads to ARM deactivation.

YawRate Condition

The function `ArmCalculateYawRateCondition` compares the sign and direction of `AyMod` and yaw rate respectively.

Since `ARM` intervention could hold or even increase the yaw rate at the previous side, it is important to avoid an `ARM` activation in the next lane, in case the yaw rate is not soon changing to the new side. If `AyMod` and the yaw rate have opposite signs, the time until the yaw rate will change sign is estimated. This time is called `YawRateZeroTransitionTime`. It holds

$$0 = \text{Yaw_Rate_Deriv} \cdot \text{YawRateZeroTransitionTime} + \text{Yaw_Rate} \quad (4.7)$$

which yields

$$\text{YawRateZeroTransitionTime} = \frac{\text{Yaw_Rate}}{\text{Yaw_Rate_Deriv}} \quad (4.8)$$

In vehicles with a brake configuration front / rear split, `ARM` uses the cross talk effect, in order to shift the wheel pressure at the front axle from one wheel to the other as soon as `ARM` wheel changes. The cross talk effect optimizes the pressure build up capability. This is considered in the YawRate Condition as well. Hence, if there is no pressure in any wheel of the front axle, the pressure build up needs more time and consequently `ARM` should activate early, even if the yaw rate needs a longer time for wheel transition. Otherwise, if there is still some pressure in one wheel, `ARM` should trigger the intervention at the new side only, when yaw rate sign matches soon to `AyMod`.

In Addition, if the yaw rate stays on the opposite side of `AyMod` for a certain time, the `Yaw_Rate_Reset` is used to deactivate `ARM`.

4.1.2 ARM Deactivation

`ARM` is deactivated, if one of the following conditions is fulfilled

- `Rollover_Crit_Expected` is reset
- `Yaw_Rate_Condition` is reset
- `Rollover_Crit_Reset`
- `Ay_Filt_Reset`
- `Yaw_Rate_Reset`

The deactivation conditions start a post run timer. As soon as the timer has finished, the `Arm_Active` Flag is reset. If the `Arm_Active` Flag is reset, the `ARM` pressure and engine commands are ramped down. Finally, the `Arm_Flag` is reset.

The primary deactivation condition for `Arm` is the `Ay_Filt_Reset` that is based on a specially filtered lateral acceleration signal (`Abs_Ay_Filt_Max`) that needs to be below a threshold for a certain time.

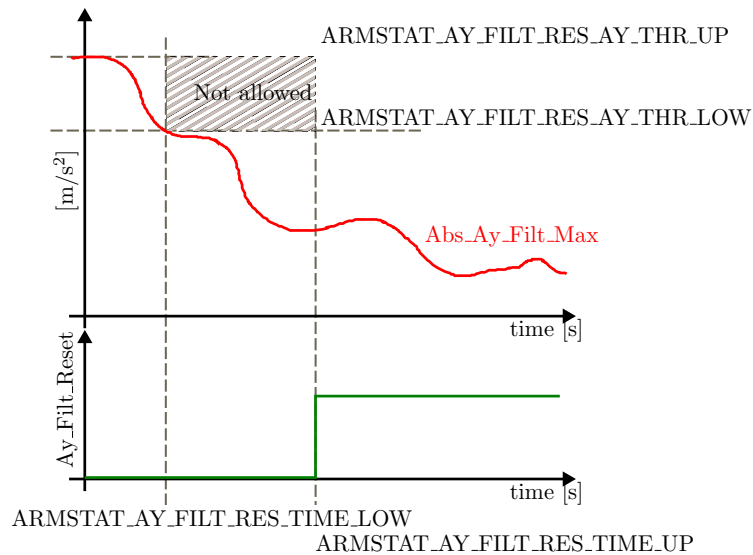


Figure 4.1: Signal condition for Ay_Filt_Reset

At first, the lateral acceleration falls below ARMSTAT_AY_FILT_RES_AY_THR_UP for the duration of ARMSTAT_AY_FILT_RES_TIME_LOW. Secondly, the lateral acceleration falls below the lower threshold ARMSTAT_AY_FILT_RES_AY_THR_LOW and causes a reset after ARMSTAT_AY_FILT_RES_TIME_UP. Finally, Ay_Filt_Reset is set.

Furthermore, the Ay_Filt_Reset is also set, when the surface friction μ falls below a threshold.

4.1.3 Vehicle Turn Direction

There are two different turn direction flags in [ARM](#). The function ArmCalculateTurnDirection determines Arm_State.Veh_Turn_Direction based on the sign of AyMod. The wheel selection for [ARM](#) intervention is derived from Arm_State.Veh_Turn_Direction.

5 PROVIDED TASKS

5.1 ArmState

Description	ArmState
Rate	Standard Arm timebase 5 - 20ms
Dependency	
SYNTAX	void ArmState (void)

6 PROVIDED SERVICES

6.0.1 Private Services

- [ArmCalculateTurnDirection](#)
- [ArmCalculateRolloverCriticality](#)
- [ArmCalculateYawRateCondition](#)

7 CONFIGURATION OPTIONS

None.

8 CONFIGURATION PROCEDURES

None.

9 TEST CASES

Test category	
Requirements	
Test Case ID	

10 CALIBRATION INFORMATION

Arm.State contains the activation and deactivation conditions. Therefore this section concentrates on the possibilities to optimize this behavior.

10.1 Activation

10.1.1 Vehicle Speed

The threshold **ARMSTAT_VELO_THR** is one of the parameters that determine if [ARM](#) intervention is allowed. If the vehicle speed is below this threshold, there is no brake intervention. A value of $\approx 45 \text{ km/h}$ is a suitable starting point. In addition, the **PRESS_CMD_SPEED_TBL**

can be used to fade in/out the brake intervention. Move **ARMSTAT_VELO_THR** between the first two supporting points (x_0 and x_1) of **PRESS_CMD_SPEED_TBL** to get the fading behavior, see Fig. 10.1. The resulting Arm_Veh.Control.Speed.Gain is directly multiplied with the **ARM** pressure request. Hence a value of zero will deactivate **ARM**.

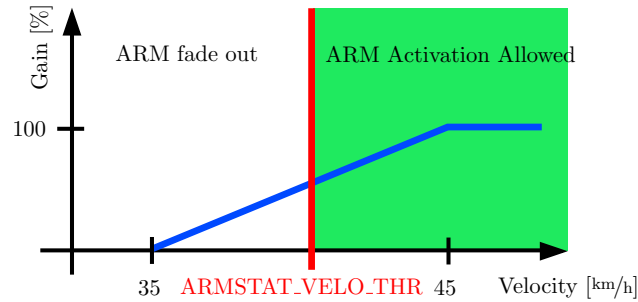


Figure 10.1: Fading out with PRESS_CMD_SPEED_TBL

10.1.2 Roll Index Flag

The roll index flag is set if a signal derived from **AyMod** is above a dynamic threshold. This threshold can be adjusted by **ROI_DEBOUNCE_GAIN_UPPER**. If **ARM** is not activating due to a missing roll index flag, one can lower this parameter.

10.1.3 Rollover Criticality

There are several parameters that influence the rollover criticality condition. First of all the thresholds for **Ay_From_Swa** and **Ay_From_Swa_Lead** are described.

Start with the base parameter **ARMAYFRSWA_THR_AY_TBL** for **Ay_From_Swa**. It determines the activation based on lateral acceleration. A suitable starting point is $\approx 14 \text{ m/s}^2$. One can adjust the threshold depending on vehicle speed by setting **ARMAYFRSWA_THR_VELO_TBL**. It is sufficient to start with 100% at first. Furthermore, so called End of Line (**EoL**) parameters can be used to adjust this threshold as well.

Continue with the base parameter **ARMAYFRSWA_LEAD_THR_AY** for **Ay_From_Swa_Lead**. A suitable starting point is $\approx 15 \text{ m/s}^2$. This threshold can be adjusted depending on vehicle speed with **ARMAYFRSWA_LEAD_THR_VELO_TBL**. It is sufficient to start with 100% at first. Furthermore, so called **EoL** parameters can be used to adjust this threshold as well.

If the signals **Ay_From_Swa** and **Ay_From_Swa_Lead** have a short delay and therefore an activation is not possible, one can use **ARMAYFRSWA_LEAD_POST_TIME** to extend the time for an activation.

If **ARM** activation is too late in second lane of a lane change maneuver, it is possible to use the **ARMSTAT_ARM_ACT_POSTRUN_TIME** and **ARM_USE_AYSWA_POSTRUN_TIMER**.

10.1.4 YawRate Condition

Start with

- **ARMSTAT_YRC_ZERO_TRAN_TIME1** = 500 ms
- **ARMSTAT_YRC_ZERO_TRAN_TIME2** = 700 ms

If the [ARM](#) intervention is triggered too early and hence the vehicle is turning in the opposite direction, one can lower the parameters above.

10.2 Deactivation

10.2.1 Ay_Filt_Reset

The deactivation can be adjusted best by Ay_Filt_Reset. The basic parameters are

- **ARMSTAT_AY_FILT_RES_AY_THR_UP**
- **ARMSTAT_AY_FILT_RES_AY_THR_LOW**
- **ARMSTAT_AY_FILT_RES_TIME_UP**
- **ARMSTAT_AY_FILT_RES_TIME_LOW**

Use the core trimming as a starting point. By lowering the ay thresholds, the [ARM](#) intervention becomes longer. Whereas by lowering the timer limits, the reset will be earlier. This has a huge impact on the robustness of [ARM](#). After these parameters were set, it is advised to keep them unchanged for the rest of the project.

11 APPENDIX